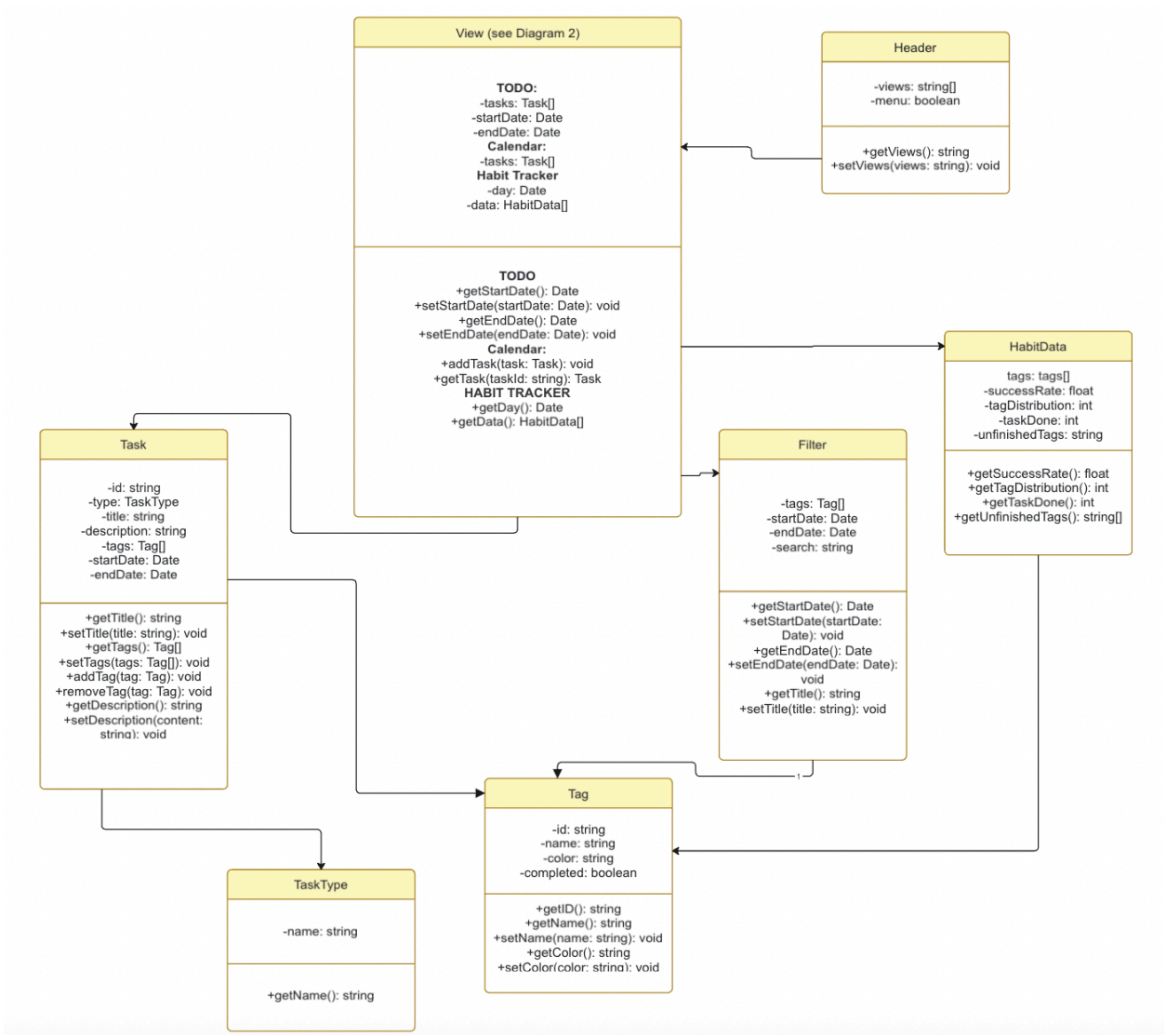
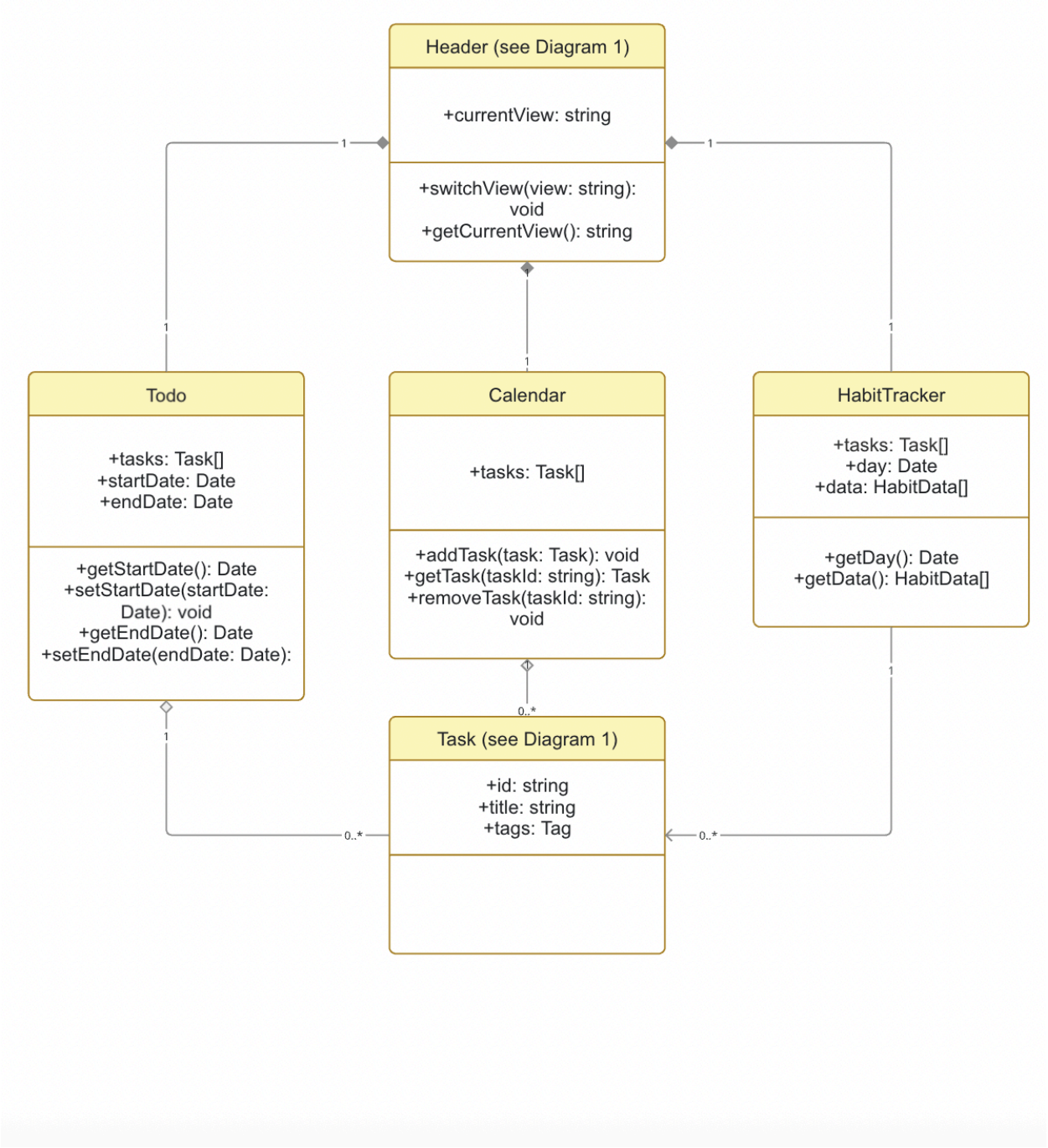
COS 420
Team Fudge
Deliverable 3
3/31

The application serves primarly to connect the user to 2 core functionalities: The Calendar and The Habit Tracker. The Calendar keeps track of what you do daily, this is information is shared with your To-do where timeless task are held. The Calendar holds Task and Tag's the primary way to keep track of your task, if you only want to see some task you can use the Filter tool to only see matching task. The Habit Tracker uses the data from how well you complete each Task and Tag to help keep the user updated on their habits either with streaks or in depth Habit Data.




Header (see Diagram 1)
+currentView: string
+switchView(view: string): void
+getCurrentView(): string
Todo
+tasks: Task[]
+startDate: Date
+endDate: Date
+getStartDate(): Date
+setStartDate(startDate: Date): void
+getEndDate(): Date
+setEndDate(endDate: Date):
Calendar
+tasks: Task[]
+addTask(task: Task): void
+getTask(taskId: string): Task
+removeTask(taskId: string): void
HabitTracker
+tasks: Task[]
+day: Date
+data: HabitData[]
+getDay(): Date
+getData(): HabitData[]
Task (see Diagram 1)
+id: string
+title: string
+tags: Tag
1
0..*

- **Task**

    - type: TaskType
    - title: string
    - content: string
    - tags: Tag[]
    - startDate: Date
    - endDate: Date
    - getTitle()
    - setTitle()
    - getTags()
    - setTags()
    - addTag()
    - removeTag()
    - getContent()
    - setContent()
    - getStartDate()
    - setStartDate()
    - getEndDate()
    - setEndDate()
- **TaskType**
    - name: string
    - getName()
- **Tag**
    - id: string
    - name: string
    - color: string
    - completed: boolean
    - getID()
    - getName()
    - setName()
    - getColor()
    - setColor()
- **Calendar**
    - tasks: Task[]
    - addTask()
    - getTask()
- **Header**
    - views: string
    - menu: boolean
    - getViews()
    - setViews()

- **ToDo**
    - tasks: task[]
    - startDate: Date
    - endDate: Date
    - getStartDate()
    - setStartDate()
    - getEndDate()
    - setEndDate()
- **HabitData**
    - successRate: floatMod
    - tagDistribution: int
    - taskDone: int
    - unfinishedTags: string[]
    - getSuccessRate()
    - getTagDistribution()
    - getTaskDone()
    - getUnfinishedTags()
- **HabitTracker**
    - day: Date
    - data: HabitData[]
    - getDay()
    - getData()
- **Filter**
    - tags: Tag[]
    - startDate: Date
    - endDate: Date
    - search: string
    - getStartDate()
    - setStartDate()
    - getEndDate()
    - setEndDate()
    - getTitle()
    - setTitle()
    - getTags()
    - setTags()
    - addTag()
    - removeTag()